DAY 03 · BEGINNER · FREE API

Prompt Templates

Write the prompt once, with named variables in it, and fill them in as many times as you like.

ChatPromptTemplate | MessagesPlaceholder | partial | {variables} | Notebook



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TEN DAYS, THREE WEEKS, ONE NOTEBOOK A DAY

The 10-Day Course Map

01

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WEEK 1 TALKING TO A MODEL

01 Your First AI Program
Ask a model, print the answer

02 Roles and Messages
The system message sets the job

03 Prompt Templates — YOU ARE HERE
One template, many inputs

YOU WILL HAVE BUILT
a script that answers anything you type into it

WEEK 2 BUILDING SOMETHING

04 Chains: The Pipe
`prompt | model | parser`

05 Structured Output
An object, not a paragraph

06 Memory
A checkpointer and a thread id

YOU WILL HAVE BUILT
a chatbot that remembers what you said a minute ago

WEEK 3 GIVING IT ABILITIES

07 Tools
Your Python, called by the model

08 Your First Agent
It decides when to call them

09 Chunks and Vectors
Find a paragraph by meaning

10 RAG with RAGWire
Citations, and a refusal

One notebook and one deck a day, **60 to 90 minutes** each. Every model used is a **free** one, and Docker only turns up on Day 9.

Why Gluing Strings Fails

GLUING STRINGS

```
"Summarise: " + email
```

in report.py

```
"Summarize this: " + body
```

in mailer.py, spelled differently

```
"in a " + tone + " tone"
```

and a silent bug when tone is empty

five places to edit, three of them wrong

ONE TEMPLATE

```
("system", "You summarise emails.
Write in a {tone} tone.")

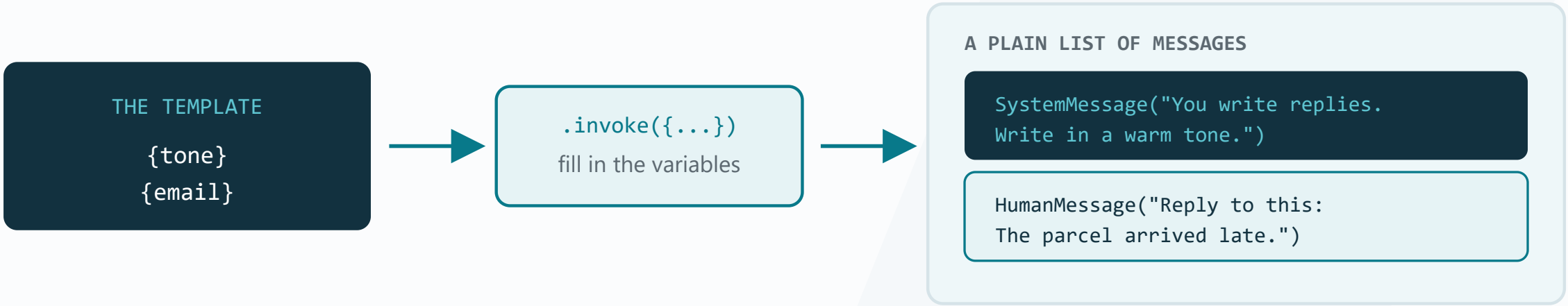
("user", "Summarise:
{email}")
```

One wording. One place to change it.

one place to edit, and it is right

A template is **the prompt with named variables in it**, and nothing has been filled in yet.

A Template Builds the Message List



exactly the list you wrote by hand on Day 2, written for you

Not a black box. You can always look at the message list it produced.

IT FAILS BEFORE ANYTHING IS SENT OR PAID FOR

A Missing Variable Raises KeyError

04

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GLUED BY HAND

```
"Write in a " + tone + " tone."
```

tone is missing, so you send:

```
"Write in a tone."
```

The model answers. Nothing raises.

You lose an hour wondering why it reads oddly.

A TEMPLATE

```
replier.invoke({"email": email})
```

```
KeyError
Input is missing variables
{'tone'}
```

Nothing was sent. Nothing was paid for.

Loud failures are cheaper than quiet ones.

MessagesPlaceholder Holds the History

THE TEMPLATE

```
("system", "concise travel assistant")
MessagesPlaceholder("history")
a slot for a whole list
("user", "{question}")
```

instructions and history stay separate



FOUR MESSAGES OUT

- System: concise travel assistant
- Human: I am flying to Delhi on Friday.
- Ai: Noted, Delhi on Friday.
- Human: What day did I say?

reply: "Friday."

The exact shape of a chatbot prompt, which is why **Day 6 plugs straight in**.

Tomorrow: Chains

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TODAY, READ INSIDE OUT

```
model.invoke(
    template.invoke({...})
).text
```

> Three calls nested in each other. You read it from the **middle** outwards.

TOMORROW, READ LEFT TO RIGHT

```
chain = template | model | parser

chain.invoke({...})
```

- One object you can pass around and reuse.
- **.batch()** for many inputs at once.
- **.stream()** so words appear as they are written.